CAPSTONE PROJECT REVIEW PROGRESS RECORD

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING | B.TECH CAPSTONE 2026–2027

Review Stage: Review 03	**Date of Review:** 06.11.2026
Project Title: Sensitivity-Guided Layer-Wise Approximate Multiplier Mapping for FPGA-Based CNN Accelerators	
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S.No.	Student Name	Student ID	Project Contribution Module	Signature
1	Velivala Keerthi	2320040083	Multiplier RTL & Error Modeling	
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1. BRIEF DESCRIPTION OF THE PROJECT

Edge artificial intelligence deployments demand energy-efficient Convolutional Neural Network (CNN) accelerators, yet silicon cost is overwhelmingly dominated by 8-bit Multiply-Accumulate (MAC) units, where an exact 8×8 multiplier consumes 68 Look-Up Tables (LUTs) on Xilinx Zynq-7020 FPGAs. While approximate computing trades controlled arithmetic noise for significant hardware savings, existing layer-wise mapping approaches rely on computationally prohibitive evolutionary search (such as ALWANN requiring 7.5 CPU-days) or costly GPU retraining loops (TRAM).

This work introduces an ultra-lightweight, single-pass sensitivity probing methodology computed in just 0.20 seconds using a 256-sample calibration probe without model retraining. By combining layer weight variance and loss gradient magnitudes, the proposed metric ranks layer-specific noise resilience, strategically steering exact arithmetic to vulnerable early feature extractors while assigning aggressive approximate multipliers to resilient deep classification layers. The entire heterogeneous accelerator pipeline is synthesized in Verilog and deployed onto a PYNQ-Z2 FPGA board via AXI4-Stream DMA, realizing up to 32.5% multiplier area reduction while preserving baseline inference classification accuracy.

2. PROGRESS ACHIEVED UP TO THIS REVIEW STAGE

- **NSGA-II Multi-Objective Benchmark & Pareto Optimality:** Executed 256-configuration NSGA-II multi-objective exploration; proved the single-pass sensitivity assignment lands directly on the Pareto knee in 0.20s, accelerating optimization by >3,000,000× over ALWANN genetic search (7.5 CPU-days) and eliminating retraining loops.
- **Complete Synthesizable Accelerator RTL Datapath:** Developed modular Verilog RTL suite (axm_8x8, mac_unit, conv_pe, layer_controller, and cnn_accelerator_top) featuring AXI4-Stream slave/master interfaces; verified against software golden model across 262,144 cycles with 100% test vector match.
- **Vivado 100 MHz Block Design Integration:** Authored automated vivado_block_design.tcl synthesizing custom IP, integrating ARM Cortex-A9 Processing System, dual AXI DMA engines (MM2S & S2MM), and high-throughput AXI Interconnect operating at 100 MHz with clean timing closure (WNS > 1.25ns).
- **PYNQ-Z2 Full-Stack Deployment:** Extracted quantized INT8 weights and biases (weights_int8.npy), built lightweight Python runtime driver (pynq_inference.py), and authored an interactive Jupyter demonstration notebook (pynq_cnn_demo.ipynb) for live edge classification.
- **IEEE Paper, 20-Slide Deck & GitHub Repository:** Authored complete 6-page IEEE conference manuscript, generated 7 publication-quality vector figures, finalized 20-slide executive presentation with embedded speaker notes, and published complete open-source codebase to GitHub.

3. STAGE 3 HARDWARE IMPLEMENTATION SUMMARY ON XILINX ZYNQ-7020 FPGA

Resource Metric	Uniform Exact (Ref)	Proposed Heterogeneous	Savings / Improvement	Target Device Margin
Multiplier Slice LUTs	10,880 LUTs	7,344 LUTs	-32.5% Multiplier Area	53,200 LUTs Available
Classification Accuracy	84.69%	84.58%	-0.11% (Preserved)	99.87% Relative Match
Exploration Latency	7.5 CPU-Days (ALWANN)	0.20 Seconds (Ours)	>3,000,000x Speedup	Single Calibration Pass
Interface & Clock	AXI4-Stream @ 100 MHz	AXI4-Stream @ 100 MHz	Full DMA Throughput	Timing Met (WNS > 1.2ns)

Evaluation Status: **[✔] Approved — All Milestone Deliverables Verified**
Next Action Focus: Final Project Viva Voce, IEEE Conference Manuscript Submission, and Open-Source Artifact Archival.

Signature of Project Guide: _____
Project Guide: Dr. Arun Nimmala | **Date:** 06.11.2026